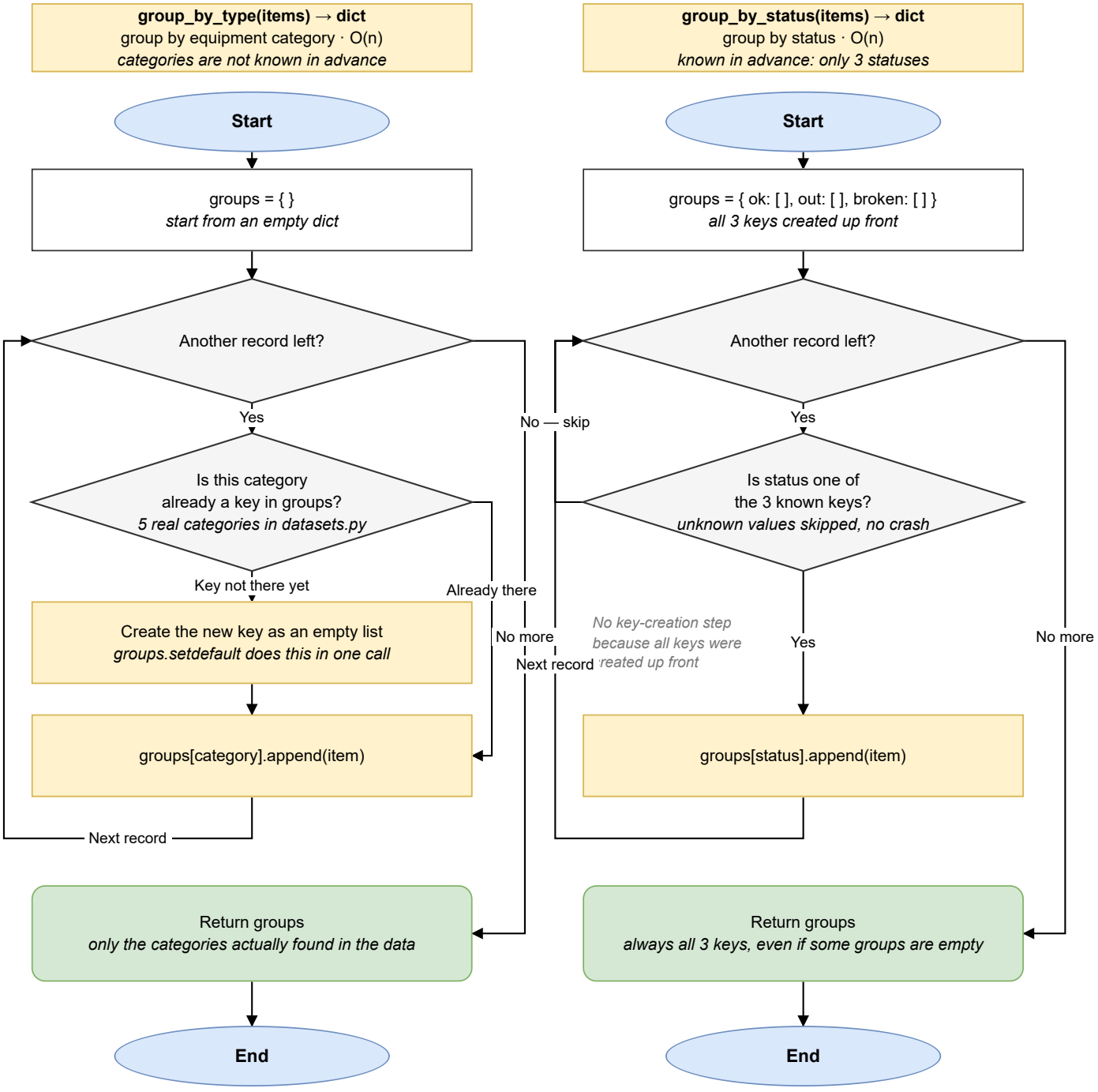
Flowchart 7 — Grouping: group_by_type and group_by_status

sort_group.py · receives items as a parameter, imports no team member's file · walks the data once like count_by_status, but stores whole records in buckets instead of only counting



Why one pass into buckets is faster than searching category by category

- The bucket method reads the data **once**, however many categories there are, so it is **O(n)**
- Searching category by category (first scan for cameras, then scan again for lenses) reads the whole data again **once per category**, so it is **O(n × number of categories)**
- The team's data set has **5 categories** (camera · lens · tripod · light · microphone), so the category-by-category method must read the data **5 times** to get the same result
- **Memory is O(n)** for both functions, because every record is stored in some bucket and none is discarded, unlike counting, which uses **O(1)**
- Unlike **Neo's count_by_status**, which keeps only **counts**, this keeps **whole records**, for when you need to know "what is there", not just "how many"
- Both functions **always return the keys they promise** — group_by_status returns all 3 keys even if some groups are empty, so callers do not have to write .get() guards everywhere, which someone would certainly forget somewhere